

BtPDR: Bluetooth and PDR-Based Indoor Fusion Localization Using Smartphones

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Abstract:

This paper presents a Bluetooth and pedestrian dead reckoning (PDR)-based indoor fusion localization approach (BtPDR) using smartphones. A Bluetooth and PDR-based indoor fusion localization approach can localize the initial position of a smartphone with the received signal strength (RSS) of Bluetooth. While a smartphone is moving, BtPDR can track its position by fusing the localization results of PDR and Bluetooth RSS. In addition, BtPDR can adaptively modify the parameters of PDR. The contributions of BtPDR include: a Bluetooth RSS-based Probabilistic Voting (BRPV) localization mechanism, a probabilistic voting-based Bluetooth RSS and PDR fusion method, and a heuristic search approach for reducing the complexity of BRPV. The experiment results in a real scene show that the average positioning error is $< 2\text{m}$, which is considered adequate for indoor location-based service applications. Moreover, compared to the traditional PDR method, BtPDR improves the location accuracy by 42.6%, on average. Compared to state-of-the-art Wireless Local Area Network (WLAN) fingerprint + PDR-based fusion indoor localization approaches, BtPDR has better positioning accuracy and does not need the same offline workload as a fingerprint algorithm.

Keywords: indoor localization; pedestrian dead reckoning; Bluetooth; received signal strength; probabilistic voting

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BtPDR: 使用智能手机进行基于蓝牙和 PDR 的室内融合定位

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抽象:

本文提出了一种基于蓝牙和行人航位推算 (PDR) 的室内融合定位方法 (BtPDR), 该方法基于智能手机。一种基于蓝牙和 PDR 的室内融合定位方法, 可以利用蓝牙的接收信号强度 (RSS) 定位智能手机的初始位置。当智能手机移动时, BtPDR 可以通过融合 PDR 和蓝牙 RSS 的定位结果来跟踪其位置。此外, BtPDR 还可以自适应地修改 PDR 的参数。BtPDR 的贡献包括: 基于蓝牙 RSS 的概率投票 (BRPV) 定位机制、基于概率投票的蓝牙 RSS 和 PDR 融合方法以及降低 BRPV 复杂性的启发式搜索方法。真实场景中的实验结果表明, 平均定位误差 < 2m, 对于室内基于位置的服务应用来说是足够的。此外, 与传统的 PDR 方法相比, BtPDR 平均将定位精度提高了 42.6%。与最先进的无线局域网 (WLAN) 指纹+基于 PDR 的融合室内定位方法相比, BtPDR 具有更好的定位精度, 并且不需要与指纹算法相同的离线工作负载。

关键词: 室内定位;行人航位推算;蓝牙;接收信号强度;概率投票

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1. Introduction

With the rapid development of the Internet and the popularity of smartphones, people's demand for location-based services (LBS) is gradually increasing [1]. Location-based services are widely used in many fields including people's work, business and military [2-3]. The key to LBS is its ability to accurately obtain the real-time position of users. Although smartphones can easily obtain an outdoor position through the Global Positioning System (GPS), they cannot easily obtain an indoor position in a building due to the occlusion of walls. Because of this, high accuracy indoor localization has become a research topic of interest in recent years.

Common methods of smartphone indoor localization are based on Received Signal Strength (RSS), for example, centroid localization [4-5], trilateration or multilateral localization [6-9] and fingerprint localization [10-13]. Centroid localization outputs the centroid of beacon nodes as a positioning result. This method is simplistic with poor positioning accuracy [4-5]. Trilateration or multilateral localization requires knowing the distances from an unknown node to at least three beacon nodes and then utilizes the distance and the position of the beacon nodes to build localization equations. Finally, it uses the maximum likelihood method [6-7], the interior point method [8], linear programming [9] and other mathematical optimization methods to calculate the position of unknown nodes. The traditional fingerprint localization approach [10-13] involves offline and online phases. In the offline phase, the approach collects RSS fingerprints of different indoor locations to create a fingerprint database. In the online phase, RSS fingerprints are compared to the fingerprints in the database to estimate the position of unknown nodes. However, traditional fingerprint localization is limited due to the necessity of a lot of offline work and the challenge of fingerprint ambiguity. Zuwei et al. [20] proposed a peer-to-peer navigation system (ppNav), avoiding the requirements of pre-deployed location services and detailed floorplans. The drawback of offline training of the fingerprint-based approach has been addressed by recent crowdsourcing-based methods in [21, 22]. To reduce RSS fingerprint ambiguity, Kaishun et al. [23] proposed a novel approach called FILA, which leverages channel state information (CSI) to build a propagation model and fingerprinting system at the receiver.

The other method of smartphone indoor localization is based on pedestrian dead reckoning (PDR). The PDR-based method employs inertial measurement unit (IMU) sensors, such as the accelerometer, gyroscope or magnetometer embedded in a smartphone, to solve indoor localization [14, 19]. Since the PDR localization method can only provide relative position information, and the localization error caused by IMU sensors may accumulate with increased walking distance [30], this method needs another assistant localization method. Therefore, the fusion localization approach, which is based on PDR and another localization method, has attracted the attention of academics [15-18]. Lyu-Han et al. [16] employed the

1. 简介

随着互联网的飞速发展和智能手机的普及, 人们的

对基于位置的服务 (LBS) 的需求正在逐渐增加[1]。基于位置的服务广泛应用于包括人们的工作、商业和军事在内的许多领域[2-3]。LBS 的关键在于它能够准确获取用户的实时位置。尽管智能手机可以通过全球定位系统 (GPS) 轻松获取室外位置, 但由于墙壁的遮挡, 它们无法轻松获取建筑物中的室内位置。正因为如此, 高精度室内定位成为近年来人们关注的研究课题。

智能手机室内定位的常用方法基于接收信号强度 (RSS), 例如质心定位[4-5]、三边定位或多边定位[6-9]和指纹定位[10-13]。质心定位将信标节点的质心输出为定位结果。这种方法简单化, 定位精度较差[4-5]。三边定位或多边定位需要知道从未知节点到至少三个信标节点的距离, 然后利用信标节点的距离和位置来构建定位方程。最后, 它使用最大似然法[6-7]、内点法[8]、线性规划[9]和其他数学优化方法来计算未知节点的位置。传统的指纹定位方法[10-13]涉及离线和在线阶段。在离线阶段, 该方法收集不同室内位置的 RSS 指纹以创建指纹数据库。在在线阶段, 将 RSS 指纹与数据库中的指纹进行比较, 以估计未知节点的位置。然而, 由于需要大量的离线工作和指纹模糊性的挑战, 传统的指纹定位受到限制。Zuwei 等[20]提出了一种点对点导航系统 (ppNav), 避免了预先部署的位置服务和详细的平面图的要求。最近基于众包的方法[21,22]解决了基于指纹的方法离线训练的缺点。为了减少 RSS 指纹模糊性, Kaishun 等[23]提出了一种称为 FILA 的新方法, 该方法利用信道状态信息 (CSI) 在接收端构建传播模型和指纹识别系统。

另一种智能手机室内定位方法是基于行人航位推算 (PDR)。基于 PDR 的方法采用惯性测量单元 (IMU) 传感器, 如嵌入智能手机中的加速度计、陀螺仪或磁力计, 来解决室内定位问题[14,19]。由于 PDR 定位方法只能提供相对位置信息, 而 IMU 传感器引起的定位误差可能会随着步行距离的增加而累积[30], 因此该方法需要另一种辅助定位方法。因此, 基于 PDR 和另一种定位方法的融合定位方法引起了学术界的关注[15-18]。Lyu-Han 等[16]采用了

maximum likelihood method to fuse fingerprint and PDR localization. Bonhyun et al. [17] employed Kalman filtering to also fuse fingerprint and PDR localization. Perttula et al. [18] used particle filter to fuse map-matching and PDR localization. However, collecting RSS fingerprints or indoor map information in [16-18] took a lot of time.

In this paper, a Bluetooth RSS and PDR-based fusion localization approach (BtPDR) for smartphone indoor localization was proposed. Although the widespread deployment of WLAN has made WLAN-based fingerprint positioning (WFP) popular, this study argues that the localization approaches, which are based on Bluetooth Low Energy (BLE), are useful in specific scenarios such as factories, indoor garages and shopping malls. Bluetooth Low Energy has the advantages of low cost, low power consumption and low penetration [24, 25]. Due to the low penetration feature of BLE, BLE beacons in different rooms or on different floors do not affect each other. Thus, BLE can achieve accurate, three-dimensional indoor localization. The Bluetooth RSS and PDR-based fusion localization approach uses Bluetooth RSS to localize the initial position of a smartphone, then employs the fusion results of Bluetooth RSS and PDR to track the trace of it. Moreover, BtPDR can adaptively modify the parameters of PDR. To summarize, the contributions of BtPDR include:

- ✧ a Bluetooth RSS-based Probabilistic Voting (BRPV) localization method to reduce the negative impact of RSS instability on localization accuracy;
- ✧ a BRPV and PDR fusion localization approach (BtPDR) to improve indoor localization accuracy;
- ✧ a heuristic search method to reduce the computational complexity of BRPV.

Experiment results in the real scene show that BtPDR can provide meter-level positioning accuracy for users with common smartphone devices, which is considered adequate for indoor LBS applications. Specifically, BtPDR's average positioning errors in three real experimental scenes were 1.86m, 1.94m and 1.58m. Compared to the traditional PDR, BtPDR improves the positioning accuracy by 42.6%, on average. In addition, compared to the state-of-the-art WLAN fingerprint + PDR-based fusion indoor localization algorithm, BtPDR has better positioning accuracy and does not need the same offline work.

2. The traditional PDR localization

The basic principle of PDR is shown in Fig. 1. First, PDR starts from a known location. Then, the heading and step length information of the walking pedestrian can be obtained by built-in sensors in a smartphone, for example, a magnetometer or an accelerometer. Using this information, the next position of the pedestrian is calculated as:

$$\begin{cases} x_{k+1} = x_k + L_k \times \sin(\theta_k) \\ y_{k+1} = y_k + L_k \times \cos(\theta_k) \end{cases} \quad (1)$$

Where (x_k, y_k) is the current location, (x_{k+1}, y_{k+1}) is the next location, θ_k is the heading,

融合指纹和 PDR 定位的最大似然法。Bonhyun 等[17]采用卡尔曼滤波来融合指纹和 PDR 定位。Perttula 等[18]使用粒子滤波来融合地图匹配和 PDR 定位。然而, 在[16-18]中收集 RSS 指纹或室内地图信息需要花费大量时间。

本文提出了一种基于蓝牙 RSS 和 PDR 的智能手机室内定位融合定位方法 (BtPDR)。尽管 WLAN 的广泛部署使得基于 WLAN 的指纹定位 (WFP) 变得流行, 但本研究认为, 基于蓝牙低功耗 (BLE) 的定位方法在工厂、室内车库和购物中心等特定场景中非常有用。低功耗蓝牙具有成本低、功耗低、穿透率低等优点[24,25]。由于 BLE 的低穿透特性, 不同房间或不同楼层的 BLE 信标互不影响。因此, BLE 可以实现准确的三维室内定位。基于蓝牙 RSS 和 PDR 的融合定位方法利用蓝牙 RSS 定位智能手机的初始位置, 然后利用蓝牙 RSS 和 PDR 的融合结果来跟踪其轨迹。此外, BtPDR 可以自适应地修改 PDR 的参数。综上所述, BtPDR 的贡献包括:

- ☒ 一种基于蓝牙 RSS 的概率投票 (BRPV) 定位方法, 以减少 RSS 不稳定性对定位精度的负面影响;
- ☒ 一种 BRPV 和 PDR 融合定位方法 (BtPDR), 以提高室内定位精度;
- ☒ 一种启发式搜索方法, 以降低 BRPV 的计算复杂度。

真实场景中的实验结果表明, BtPDR 可以为使用普通智能手机设备的用户提供米级的定位精度, 这对于室内 LBS 应用来说被认为是足够的。具体来说, BtPDR 在三个真实实验场景中的平均定位误差分别为 1.86m、1.94m 和 1.58m。与传统 PDR 相比, BtPDR 平均提高了 42.6% 的定位精度。此外, 与最先进的 WLAN 指纹+基于 PDR 的融合室内定位算法相比, BtPDR 具有更好的定位精度, 不需要相同的离线工作。

2. 传统的 PDR 国产化

PDR 的基本原理如图 1 所示。首先, PDR 从已知位置开始。

然后, 可以通过智能手机中的内置传感器 (例如磁力计或加速度计) 获取行人的航向和步长信息。使用这些信息, 行人的下一个位置计算如下:

$$\begin{aligned} x_{k+1} &= x_k + L_k \times \cos(\theta_k) \\ y_{k+1} &= y_k + L_k \times \sin(\theta_k) \end{aligned} \quad (1)$$

其中 (x_k, y_k) 是当前位置, (x_{k+1}, y_{k+1}) 是下一个位置, θ 是标题,

and L_k is the step length.

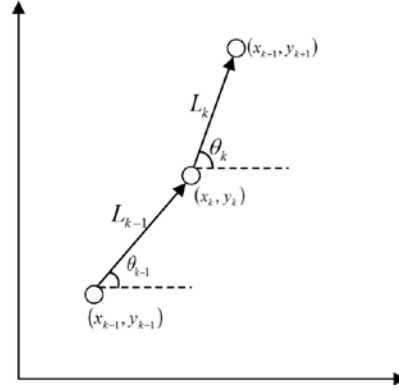


Fig. 1. PDR algorithm schematic

When pedestrians are moving, an acceleration waveform, collected by the built-in acceleration sensor, is created, as shown in **Fig. 2**. Each cycle waveform in **Fig. 2** represents a forward step. After filtering the high frequency components and limiting amplitude, the acceleration waveform is shown in **Fig. 3**. By using the zero-crossing detection method in [14], the number of walking steps was counted.

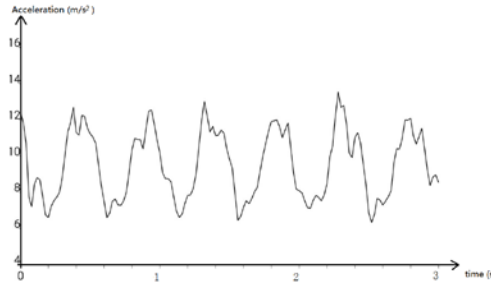


Fig. 2. Acceleration waveform during normal walking

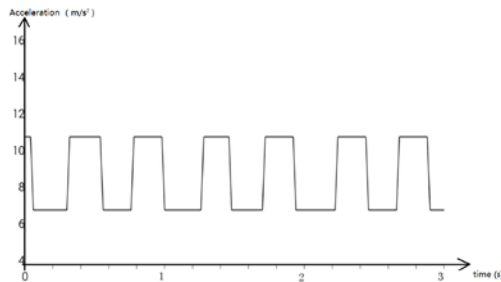


Fig. 3. Acceleration waveform after filtering

To estimate step size, under the condition of an acceleration sensor fixed in shoes, a step size formula was proposed in [14] as follows:

$$L = K * \sqrt[4]{a_{\max} - a_{\min}} \quad (2)$$

L 是步长。

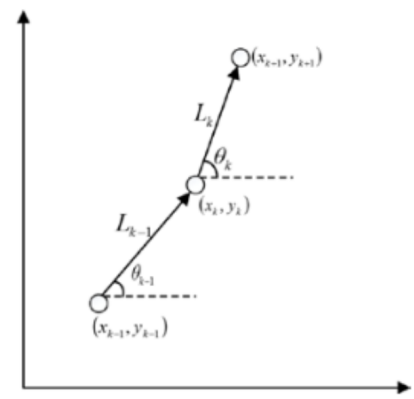


图 1.PDR 算法示意图

当行人移动时，会创建一个由内置加速度传感器收集的加速度波形，如图 2 所示。图 2 中的每个循环波形代表一个前进的步长。滤波高频分量并限制幅度后，加速度波形如图 3 所示。通过使用[14]中的过零检测方法，计算了行走的步数。

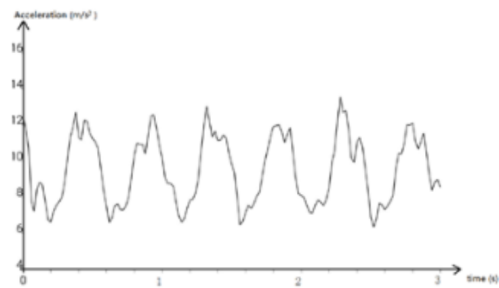


图 2.正常行走时的加速度波形

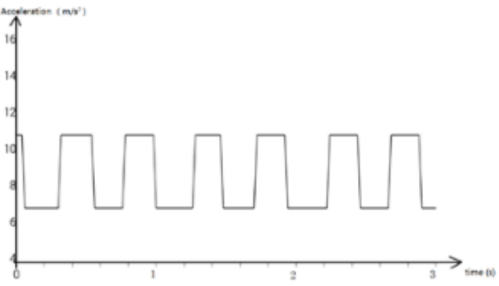


图 3.滤波后的加速度波形

为了估计步长，在固定在鞋子中的加速度传感器的情况下，在[14]中提出了步长公式如下：

$$L = K \cdot \sqrt[4]{a_{\text{麦克斯}} \cdot a_{\text{均}}}$$

(2)

Where K is a constant value, belonging to $[0.2, 0.5]$, and is related to the gender, height and weight of the pedestrian and $a_{\max}$ and $a_{\min}$ denote the maximum and minimum acceleration values in a step obtained by the acceleration sensor. However, BtPDR uses built-in sensors and assumes that the pedestrian holds a smartphone in their hand. Therefore, for this experiment, the step size formula should be different than in [14]. According to the experiment results, BtPDR employs a step size formula as follows:

$$L = K * \sqrt[4]{(a_{\max} - a_{\min}) + 2} \quad (3)$$

The main drawbacks of traditional PDR are as follows: 1) it is unable to determine the initial location; 2) it uses acceleration changes to estimate step size and the estimation error varies with the individual and 3) it is relative localization and cannot eliminate cumulative errors, hence, its cumulative error grows with increased walking distance.

3. The proposed BRPV localization

To overcome the shortcomings of traditional PDR, a BRPV to assist PDR to realize high-precision indoor localization was proposed.

3.1 The RSS ranging model

Due to the weak penetrability of Bluetooth radio, a smartphone and Bluetooth beacon are usually in the same enclosed space to communicate with each other. Therefore, the signal propagation path loss between the smartphone and Bluetooth beacon can be modeled with a log-normal distribution as shown in (4).

$$P_{r,db}(d) = P_{r,db}(d_0) - 10 \times n_p \times \log(d / d_0) + X \quad (4)$$

Where d_0 is the reference distance; d is the distance between the smartphone and Bluetooth beacon; $P_{r,db}(d)$ is the path loss at d_0 distance to the Bluetooth beacon; n_p is the path loss constant, usually equaling 2~6 and X is a Gaussian random variable that obeys $N(0, \sigma^2)$. Then, according to (4), the distance d can be obtained as follows:

$$d = d_0 \times 10^{(P_{r,db}(d_0) - P_{r,db}(d)) / (10 \times n_p)} \quad (5)$$

However, in the actual application, due to noise interference (e.g., multipath and shadow effect), $P_{r,db}(d)$ has large fluctuations, meaning that the estimated distance may be far from the real distance. Therefore, traditional RSS range-based multilateral localization may produce large localization errors at times.

3.2 The proposed BRPV

To tolerate RSS instability, BRPV does not directly use RSS to compute the distance between the smartphone and Bluetooth beacon; rather, it uses RSS to compute the

式中, K 为常数值, 属于 $[0.2, 0.5]$, 与行人的性别、身高和体重有关, 并且表示加速度传感器获得的步长中的最大和最小加速度值。然而, BtPDR 使用内置传感器, 并假设行人手中拿着智能手机。因此, 对于本实验, 步长公式应与[14]中不同。根据实验结果, BtPDR 采用的步长公式如下:

$$L = K * \sqrt{(a_{\text{max}} - a_{\text{min}})^2} + 2 \quad (3)$$

传统 PDR 的主要缺点是: 1) 无法确定初始位置; 2) 利用加速度变化来估计步长, 估计误差因人而异, 3) 是相对定位, 无法消除累积误差, 因此其累积误差随着步行距离的增加而增加。

3. 拟议的 BRPV 国产化

针对传统 PDR 的不足, 提出了一种 BRPV 辅助 PDR 实现高精度室内定位。

3.1 RSS 测距模型由于蓝牙无线电的穿透性较弱, 智能手机和蓝牙信标通常在同一封闭空间内相互通信。因此, 智能手机和蓝牙信标之间的信号传播路径损耗可以用对数正态分布建模, 如 (4) 所示。

$$P_{r,db}(d) = P_{r,db}(d_0) - 10 \times n_p \times \log_{10}(d/d_0) + X \quad (4)$$

式中 d_0 为参考距离; d 是智能手机与蓝牙信标之间的距离; $P(d)$ 是 d 距离蓝牙信标处的路径损耗; n 是路径损耗常数, 通常等于 2~6, X 是服从 $N(0, \sigma)$ 的高斯随机变量。那么, 根据 (4), 可以得到距离 d 如下:

$$d = d_0 \times 10^{(P_{r,db}(d_0) - P_{r,db}(d)) / (10 \times n_p)} \quad (5)$$

然而, 在实际应用中, 由于噪声干扰 (如多径和阴影效应), $P(d)$ 具有较大的波动, 这意味着估计的距离可能与实际距离相去甚远。因此, 传统的基于 RSS 范围的多边定位有时可能会产生较大的定位误差。

3.2 所提出的 BRPV为了容忍 RSS 的不稳定性, BRPV 不直接使用 RSS 来计算智能手机和蓝牙信标之间的距离; 相反, 它使用 RSS 来计算

probability of a smartphone's true location in different areas. Specifically, BRPV uses the Bluetooth beacon as a central point to obtain multiple concentric rings then calculates the probability that the real position of the smartphone falls into each concentric ring. When there are multiple Bluetooth beacons, the concentric rings of different Bluetooth beacons will intersect to create multiple areas. Then the optimal area will be selected by using the probabilistic voting method on each Bluetooth beacon. Finally, the centroid of the optimal area will be regarded as the location result. Advantages of BRPV are that it has outstanding tolerance to RSS instability and it can output localization results with a degree of reliability.

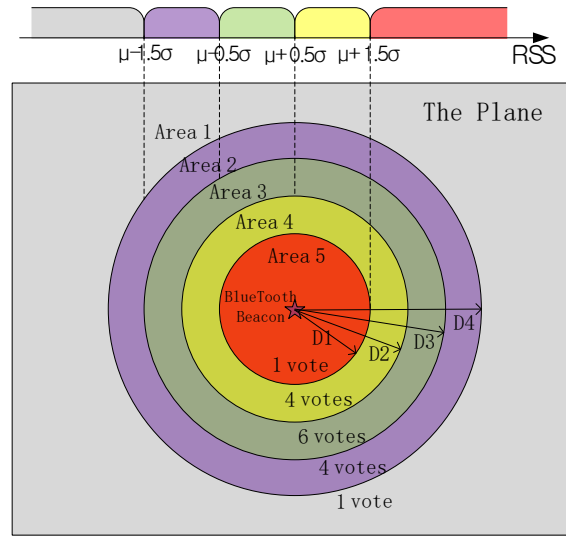


Fig. 4. RSS mapping distance

The articles in [26-28] point out that an RSS at a fixed position generally obeys the Gaussian distribution of $N(\mu, \sigma^2)$. To make a tradeoff between localization accuracy and RSS instability tolerance, BRPV defines four concentric rings, as shown in Fig. 4. The radiuses of these four rings are $D1$, $D2$, $D3$ and $D4$. These four rings divide the plane into five areas and the corresponding RSS ranges are $(-\infty, \mu-1.5\sigma]$, $(\mu-1.5\sigma, \mu-0.5\sigma]$, $(\mu-0.5\sigma, \mu+0.5\sigma]$, $(\mu+0.5\sigma, \mu+1.5\sigma]$ and $(\mu+1.5\sigma, +\infty)$, where μ is the measured RSS and σ is a pre-determined standard deviation. That is to say, $D1$, $D2$, $D3$ and $D4$ are obtained by (5), with $P_{r,dB}(d)$ equaling $\mu+1.5\sigma$, $\mu+0.5\sigma$, $\mu-0.5\sigma$ and $\mu-1.5\sigma$, respectively.

After the smartphone receives the broadcast of the Bluetooth beacon, the probability of its real position in these five areas can be calculated by (6), where dis_{true} is the true distance between the smartphone and the Bluetooth beacon. Then, the number of votes of each area can be decided according to corresponding probability of each area, as shown in Fig. 4. The probability of areas 5 and 1 is 0.067 and they are assigned one vote. The probability of areas 4 and 2 is 0.242, about 3.6 times more than that of areas 5 and 1, hence, they are assigned four votes. The probability of area 3 is 0.383, about 5.73 times more than that of areas 5 and

智能手机在不同区域的真实位置的概率。具体来说, BRPV 以蓝牙信标为中心点, 获取多个同心环, 然后计算智能手机的真实位置落入每个同心环的概率。当有多个蓝牙信标时, 不同蓝牙信标的同心环会相交, 形成多个区域。然后, 通过在每个蓝牙信标上使用概率投票方法来选择最优区域。最后, 将最优区域的质心视为定位结果。BRPV 的优点是对 RSS 不稳定性具有出色的容忍度, 并且可以输出具有一定可靠性的定位结果。

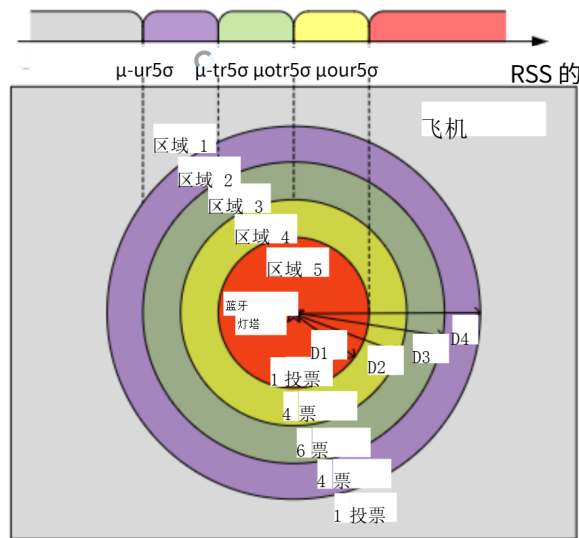


图 4. RSS 映射距离

[26-28]中的文章指出, 处于固定位置的 RSS 通常服从 $N(\mu, \sigma)$ 的高斯分布。为了在定位精度和 RSS 不稳定性容忍之间进行权衡, BRPV 定义了四个同心环, 如图 4 所示。这四个环的半径是 $D1$ 、 $D2$ 、 $D3$ 和 $D4$ 。这四个环将平面划分为五个区域, 相应的 RSS 范围为 $(-\infty, \mu - 1.5\sigma]$ 、 $(\mu - 1.5\sigma, \mu - 0.5\sigma]$ 、 $(\mu - 0.5\sigma, \mu + 0.5\sigma]$ 、 $(\mu + 0.5\sigma, \mu + 1.5\sigma]$ 和 $(\mu + 1.5\sigma, +\infty)$, 其中 μ 是测量的 RSS, σ 是预先确定的标准差。也就是说, $D1$ 、 $D2$ 、 $D3$ 和 $D4$ 由 (5) 得到, 其中 $P(d)$ 分别等于 $\mu + 1.5\sigma$ 、 $\mu + 0.5\sigma$ 、 $\mu - 0.5\sigma$ 和 $\mu - 1.5\sigma$ 。

智能手机接收到蓝牙信标的广播后, 其在这五个区域的真实位置的概率可以通过 (6) 计算, 其中 dis 是智能手机与蓝牙信标之间的真实距离。然后, 可以根据每个区域对应的概率来决定每个区域的票数, 如图 4 所示。区域 5 和 1 的概率为 0.067, 它们被分配一票。区域 4 和 2 的概率为 0.242, 大约是区域 5 和 1 的 3.6 倍, 因此, 它们被分配了四票。区域 3 的概率为 0.383, 大约是区域 5 和区域 5 的 5.73 倍。

1, so area 3 is assigned six votes.

$$P = \begin{cases} P(\text{the smartphone is in area 1}) = P(dis_{ture} > D_4) = \phi(\mu - 1.5\sigma) = 0.067 \\ P(\text{the smartphone is in area 2}) = P(D_3 < dis_{ture} < D_4) = \phi(\mu - 0.5\sigma) - \phi(\mu - 1.5\sigma) = 0.242 \\ P(\text{the smartphone is in area 3}) = P(D_2 < dis_{ture} < D_3) = \phi(\mu + 0.5\sigma) - \phi(\mu - 0.5\sigma) = 0.383 \\ P(\text{the smartphone is in area 4}) = P(D_1 < dis_{ture} < D_2) = \phi(\mu + 1.5\sigma) - \phi(\mu + 0.5\sigma) = 0.242 \\ P(\text{the smartphone is in area 5}) = P(dis_{ture} < D_1) = 1 - \phi(\mu + 1.5\sigma) = 0.067 \end{cases} \quad (6)$$

The specific localization process of BRPV is as follows:

Assume that the received RSS vector of the smartphone is $\overrightarrow{RSS} = (rss_k^1, rss_k^2, \dots, rss_k^N)$, where rss_k^i denotes the RSS of the i -th Bluetooth beacon in the k -th step. According to rss_k^i , the probability voting areas of the i -th Bluetooth beacon can be computed. An example of probability voting areas with three Bluetooth beacons is shown in Fig. 5. From Fig. 5 it can be seen that the probability voting areas of different Bluetooth beacons overlap each other, dividing the plane into more subareas. The number of votes for each subarea is equal to the sum of the number of votes polled by each Bluetooth beacon. The centroid of the subarea with the most votes is the final localization result. For example, if the shadowed subarea in Fig. 5 has the most votes, then the final localization result is its centroid.

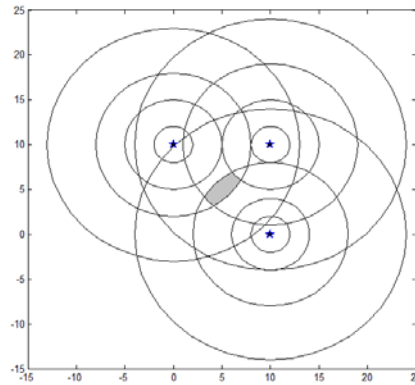


Fig. 5. Probability voting areas with three Bluetooth beacons

3.3 A full search method for BRPV

When there are many Bluetooth beacon nodes, the plane is divided into a lot of voting subareas, as shown in Fig. 5. In this case, it is very difficult to compute the boundary of each subarea and its centroid. Focusing on this problem, a simple, approximate calculation method, called a full search, was employed to derive the location with the most number of votes.

In the full search, a grid lattice to represent a subarea was employed, as shown in Fig. 6. For each grid point, votes were computed according to (7). The grid point with the most number of votes was used as the localization results of BRPV. If there were multiple grid points with the same number of votes, their averaged value was used as the localization results of BRPV. While the full search method has high computation complexity if the grid

1, 因此区域 3 被分配了六票。

$$\begin{aligned}
 P(\text{智能手机位于区域 } P) &= \frac{P(\text{智能手机位于区域 } P \mid \text{智能手机位于区域 } P)}{P(\text{智能手机位于区域 } P)} = \frac{0.067}{0.067} = 1.0 \\
 P(\text{智能手机位于区域 } P \mid \text{智能手机位于区域 } P) &= \frac{P(\text{智能手机位于区域 } P \mid \text{智能手机位于区域 } P)}{P(\text{智能手机位于区域 } P)} = \frac{0.067}{0.067} = 1.0 \\
 P(\text{智能手机位于区域 } P \mid \text{智能手机位于区域 } P) &= \frac{P(\text{智能手机位于区域 } P \mid \text{智能手机位于区域 } P)}{P(\text{智能手机位于区域 } P)} = \frac{0.067}{0.067} = 1.0 \\
 P(\text{智能手机位于区域 } P \mid \text{智能手机位于区域 } P) &= \frac{P(\text{智能手机位于区域 } P \mid \text{智能手机位于区域 } P)}{P(\text{智能手机位于区域 } P)} = \frac{0.067}{0.067} = 1.0 \\
 P(\text{智能手机位于区域 } P \mid \text{智能手机位于区域 } P) &= \frac{P(\text{智能手机位于区域 } P \mid \text{智能手机位于区域 } P)}{P(\text{智能手机位于区域 } P)} = \frac{0.067}{0.067} = 1.0 \\
 P(\text{智能手机位于区域 } P \mid \text{智能手机位于区域 } P) &= \frac{P(\text{智能手机位于区域 } P \mid \text{智能手机位于区域 } P)}{P(\text{智能手机位于区域 } P)} = \frac{0.067}{0.067} = 1.0
 \end{aligned}$$

BRPV 的具体定位过程如下: 假设智能手机接收到的 RSS 向量是

$(RSS_1, RSS_2, \dots, RSS_n)$

哪里 RSS_i 表示第 k 步中第 i 个蓝牙信标的 RSS。根据 RSS_i , 可以计算出第 i 个蓝牙信标的概率投票区域。图 5 显示了具有三个蓝牙信标的概率投票区域的示例。从图 5 可以看出, 不同蓝牙信标的概率投票区域相互重叠, 将平面划分为更多的子区域。每个子区域的票数等于每个蓝牙信标轮询的票数之和。得票最多的子区域的质心是最终的定位结果。例如, 如果图 5 中的阴影子区域得票最多, 则最终的定位结果就是其质心。

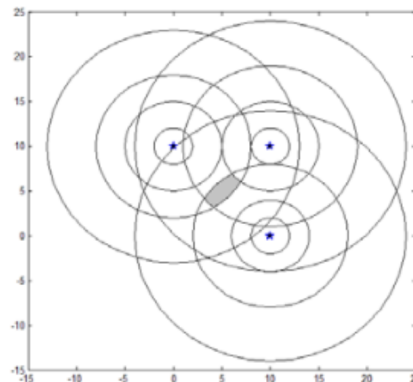


图 5. 具有三个蓝牙信标的概率投票区域

3.3 BRPV 的全搜索方法当蓝牙信标节点较多时, 平面被划分为多个投票子区域, 如图 5 所示。在这种情况下, 计算每个子区域的边界及其质心非常困难。针对这个问题, 采用了一种简单的近似计算方法, 称为全搜索, 以推导出得票数最多的位置。

在完整的搜索中, 使用了表示子区域的网格晶格, 如图 6 所示。

对于每个网格点, 根据 (7) 计算投票结果。投票数最多的网格点作为 BRPV 的定位结果。如果存在多个具有相同投票数的网格点, 则使用其平均值作为 BRPV 的定位结果。而全搜索方法的计算复杂度较高, 如果网格

size is small, a smaller grid size can help to improve localization accuracy as the space is finely described. In this paper, a grid size of 0.1 m was chosen.

$$(\mathbf{x}_{rss}, y_{rss}) = \arg \max_{(x,y)} V \equiv \sum_{i=1}^m v_i(x, y) \quad (7)$$

Where m is the number of Bluetooth beacons, $v_i(x, y)$ is the number of votes polled by the i -th Bluetooth beacon for the position (x, y) , V is the total number of votes polled by all Bluetooth beacons, and $(\mathbf{x}_{rss}, y_{rss})$ is the localization results of the full search method.

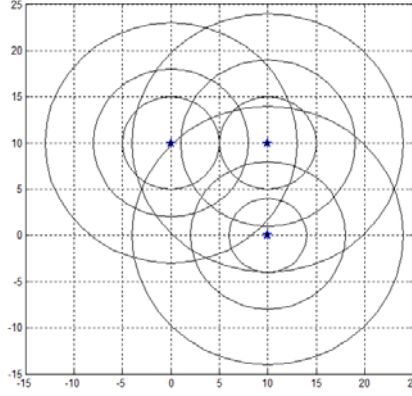


Fig. 6. The proposed full search scheme of BRPV

3.4 A heuristic search method for BRPV

Although the idea of the full search method is simple, and its localization accuracy is high, its complexity is also high, which limits its application with smartphones. Therefore, a heuristic search scheme for BRPV was proposed. The specific steps are as follows:

Step 1: Determine the initial search location $\mathbf{x}_{rss}(0) = \mathbf{x}_{pdr} = (x_{pdr}, y_{pdr})$. BRPV employs $\mathbf{x}_{pdr}$, the localization results of PDR, as the initial search position.

Step 2: Take N equal interval points from the circle whose center is $\mathbf{x}_{rss}(0)$ and radius is $search_step$, which is shown in **Fig. 7**. The coordinates of these N search points are as follows:

$$\begin{aligned} \mathbf{x}e_k &= \mathbf{x}e_0 + search_step \times [\cos \theta_k, \sin \theta_k], \\ \theta_k &= (k-1) * \frac{2\pi}{N}, \quad k = 1, 2, \dots, N \end{aligned} \quad (8)$$

Step 3: Calculate the value of the objective function, defined in (9), at the above $N+1$ search points (including the circle center).

$$f(\mathbf{x}e_k) = \sum_{i=1}^m V_i(\mathbf{x}e_k), \quad k = 0, 1, \dots, N \quad (9)$$

Step 4: Assume that there are M positions which have the same maximum value $\mathbf{x}_{rss_opt}$ in the current search iteration. Then, BRPV updates $\mathbf{x}_{rss}(0)$ and $search_step$ according to (11)

如果网格尺寸较小, 全搜索方法的计算复杂度较高, 但较小的网格尺寸有助于提高定位精度, 因为空间被精细描述。本文选择了 0.1 m 的网格尺寸。

$$(X_{RSS}, Y_{RSS}) = \underset{(x,y)}{\text{argmax}} \sum_{i=1}^m V_i(x,y) \quad (7)$$

其中 V_i 是第 i 个蓝牙信标在位置 (x, y) 投票的票数, V 是所有蓝牙信标投票的总票数, (x, y) 是第 i 个蓝牙信标的位置。

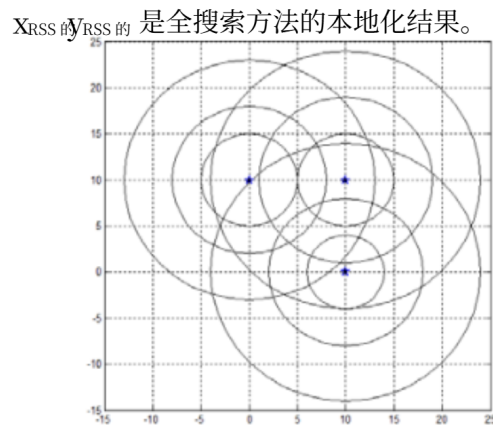


图 6.BRPV 全搜索方案

3.4 BRPV 的启发式搜索方法全量搜索方法虽然思想简单, 定位精度高, 但复杂度也高, 限制了其在智能手机上的应用。因此, 提出了一种 BRPV 的启发式搜索方案。具体步骤如下:

步骤 1: 确定初始搜索位置 $x_e(0) = x_e = (x, y)$ 。BRPV 采用 PDR 的定位结果 x_e 作为初始搜索位置。步骤 2: 从中心为 $x_e(0)$ 且半径为 search_step 的圆中取 N 个等间隔点, 如图 7 所示。这 N 个搜索点的坐标如下:

$$\begin{aligned} x_k &= x_e + \text{search_step} \times \cos \theta_k \\ y_k &= y_e + \text{search_step} \times \sin \theta_k, \\ \theta_k &= (k-1) \times \frac{2\pi}{N}, \quad k=1, 2, \dots, N \end{aligned} \quad (8)$$

步骤 3: 计算 (9) 中定义的目标函数在上述 $N+1$ 个搜索点 (包括圆心) 处的值。

$$f(x_k) = \sum_{i=1}^m V_i(x_k) / N, \quad k=0, 1, \dots, N \quad (9)$$

第 4 步: 假设在当前搜索迭代中存在 M 个位置具有相同的最大值 $x_{\text{erss_opt}}$ 。然后, BRPV 根据 (11) 更新 $x_e(0)$ 和 search_step

and (12) for the next search iteration.

$$xe_{rss_opt} = \max_{k=0,1,\dots,N} \{f(xe_k)\} \quad (10)$$

$$xe_{rss}(0) = \frac{1}{M} \sum_{i=1}^M xe_{rss_opt}(i) \quad (11)$$

$$seach_step = \alpha * seach_step \quad (12)$$

Step 5: Repeat steps 2~4 until $seach_step$ is less than $stop_th$. The localization results of BRPV are the last updated $xe_{rss}(0)$.

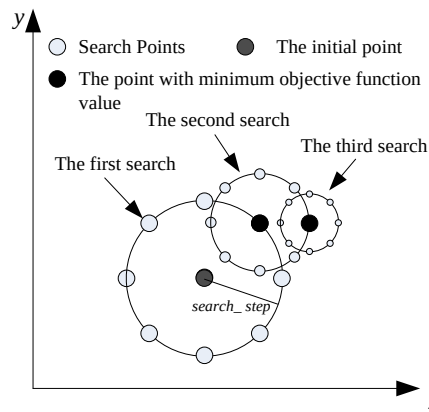


Fig. 7. The proposed heuristic search scheme of BRPV

In the previous BRPV, α is the convergence factor, which is used to control the convergence speed and, in theory, belongs to $(0, 1)$ and $stop_th$ is the search stopping threshold. The localization accuracy and complexity of BRPV are mainly determined by α and $stop_th$. Generally, the smaller $stop_th$ is and the larger α is, the higher the localization accuracy and complexity will be.

In the simulation experiment, $N = 16$, $search_step = 5$, $stop_th = 0.1$, and $\alpha = 0.8$, meaning that BRPV needed to search approximately 306 points in each localization calculation. However, if the full search method with 0.1m grid size described in Section 3.3 had been employed, BRPV would have needed to search approximately 15,000 points in each localization calculation with a location area of $6m \times 25m$. The number of search points of the proposed heuristic search method is only 2.04% that of the full search method.

4. The proposed BtPDR localization

4.1 The specific flow of the BtPDR

The traditional PDR can neither determine initial position nor eliminate cumulative error

(12) 用于下一次搜索迭代。

$$x_{RSS_选择} = \max_{k=0,1,2,...,N} \{f(x_k)\}$$

(10)

$$x_{RSS}(0) = \frac{1}{M} \sum_{i=1}^M x_{RSS_选择}(i)$$

(11)

$$查看_步 = a * 查看_步$$

(12)

步骤 5: 重复步骤 2~4, 直到 search_step 小于 stop_th。BRPV 的定位结果是最后更新的 $x_e(0)$ 。

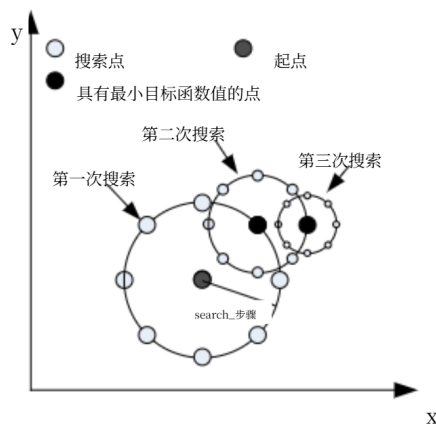


图 7.BRPV 启发式搜索方案

在之前的 BRPV 中, α 是收敛因子, 用于控制收敛速度, 理论上属于 (0,1), $stop_th$ 是搜索停止阈值。BRPV 的定位精度和复杂度主要由 a 和 $stop_th$ 决定。一般来说, $stop_th$ 越小, ra 越大, 定位精度和复杂度就越高。

在仿真实验中, $N = 16$, $search_step = 5$, $stop_th = 0.1$, $\alpha = 0.8$, 这意味着 BRPV 在每次定位计算中需要搜索大约 306 个点。然而, 如果采用第 3.3 节中描述的网格大小为 0.1m 的完整搜索方法, BRPV 在每次定位计算中需要搜索大约 15,000 个点, 位置面积为 $6m \times 25m$ 。所提出的启发式搜索方法的搜索点数量仅为完整搜索方法的 2.04%。

4. 提出的 BtPDR 本地化

4.1 BtPDR 的具体流量传统 PDR 既不能确定初始位置, 也不能消除累积误差

and estimate the step size accurately, and the localization accuracy of the traditional RSS range-based localization is low when each step receives few RSS samples. To solve the presented problems, a Bluetooth RSS and PDR-based fusion localization approach (BtPDR) for smartphone indoor localization and tracking systems was proposed. First, the initial position of a smartphone is localized with the proposed BRPV algorithm. Second, while the smartphone is moving, it is tracked with the traditional PDR and BRPV at the same time. Last, the proposed BtPDR can intelligently fuse PDR and BRPV localization results, adaptively modifying the parameters of PDR. The overall system architecture of BtPDR is shown in Fig. 8.

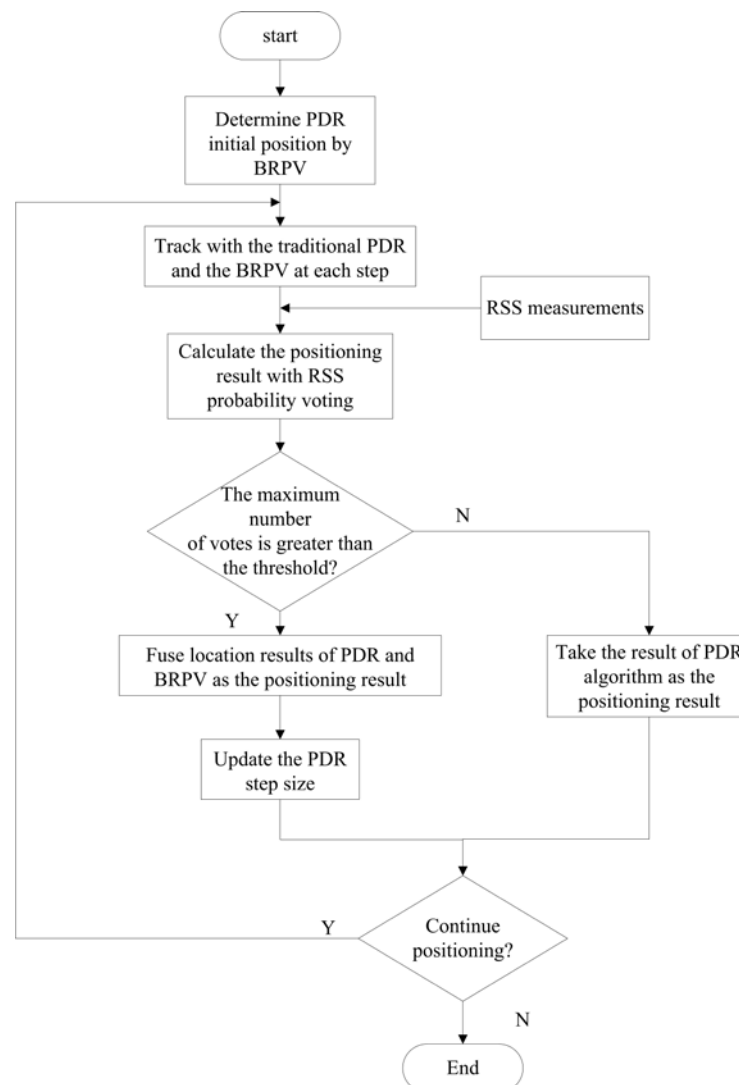


Fig. 8. Overall system architecture of BtPDR

并准确估计步长, 当每一步接收的 RSS 样本较少时, 传统 RSS 基于范围的定位的定位精度较低。为了解决所提出的问题, 提出了一种基于蓝牙 RSS 和 PDR 的智能手机室内定位和跟踪系统融合定位方法 (BtPDR)。首先, 使用所提出的 BRPV 算法对智能手机的初始位置进行定位。其次, 在智能手机移动的同时, 同时使用传统的 PDR 和 BRPV 进行跟踪。最后, 所提出的 BtPDR 可以智能地融合 PDR 和 BRPV 定位结果, 自适应地修改 PDR 的参数。BtPDR 的整体系统架构如图 8 所示。

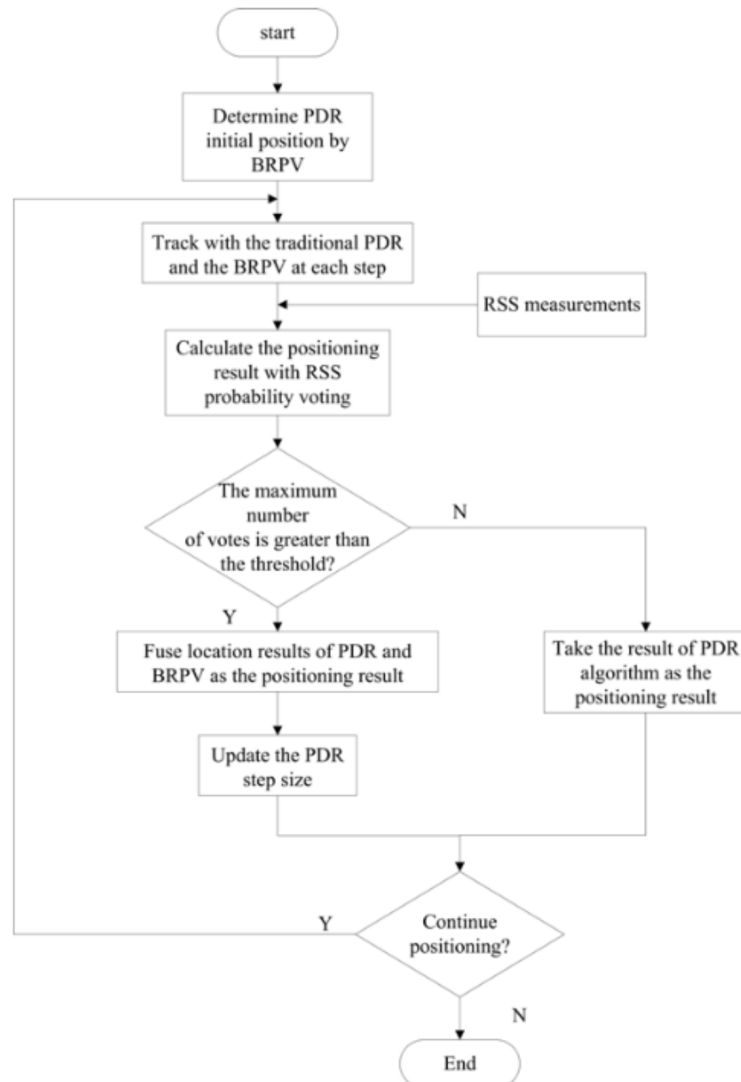


图 8. BtPDR 整体系统架构

4.2 Fusion rules of BtPDR

During the smartphone tracking process, after $\mathbf{x}e_{pdr}$, the positioning results of PDR, and $\mathbf{x}e_{rss}$, the positioning results of BRPV, are derived, the next key issue is how to fuse $\mathbf{x}e_{pdr}$ and $\mathbf{x}e_{rss}$. As mentioned, $\mathbf{x}e_{rss}$ is easily affected by the fluctuations of RSS and the number of votes at $\mathbf{x}e_{rss}$ reflects the probability that the actual position being in a region; the greater the number of votes, the greater the probability of the real location being in the region. Therefore, to reduce the influence of the RSS fluctuation, BtPDR discards $\mathbf{x}e_{rss}$ with the number of votes less than or equal to a threshold β . Only when the number of votes V_{rss} of BRPV is greater than β , then $\mathbf{x}e_{rss}$ is believed to be reliable. When the results of PDR and BRPV need to be fused, the fusion is based on:

$$\mathbf{x}e = \begin{cases} \frac{V_{pdr}}{V_{rss} + V_{pdr}} \cdot \mathbf{x}e_{pdr} + \frac{V_{rss}}{V_{rss} + V_{pdr}} \cdot \mathbf{x}e_{rss} & V_{rss} > \max\{\beta, V_{pdr}\} \\ \mathbf{x}e_{pdr} & otherwise \end{cases} \quad (13)$$

Where $\mathbf{x}e$ is the final fusion localization results and V_{rss} and V_{pdr} are the number of votes by probabilistic voting at $\mathbf{x}e_{rss}$ and $\mathbf{x}e_{pdr}$, respectively. It can be seen in (13) that the weighted value of $\mathbf{x}e$ is the number of votes of $\mathbf{x}e_{pdr}$ and $\mathbf{x}e_{rss}$. The idea behind the weighted fusion method is to avoid the emergence of large errors by limiting the correction range of positions, making the fusion results more reliable. Moreover, the initial position localized by BRPV only affects the accuracy at the beginning. After BtPDR fuses the localization results of PDR and BRPV several times, the localization accuracy will converge and be independent of the initial position.

4.3 The parameter update of BtPDR

After BtPDR fuses the localization results of PDR and BRPV, it needs to adaptively update the step size of PDR. In this study, it was assumed that (x_k, y_k) are the coordinates of the k -th fusion results and (x_{pdr}, y_{pdr}) are the coordinates of the current PDR. Then the step size correction ratio *rate* can be computed:

$$rate = \frac{\sqrt{(x_{k-1} - x_k)^2 + (y_{k-1} - y_k)^2}}{\sqrt{(x_{k-1} - x_{pdr})^2 + (y_{k-1} - y_{pdr})^2}} \quad (14)$$

When the *rate* is greater than 0.6 and less than 1.5, it is an effective correction. Next, BtPDR updates the K in (3) as follows:

$$K_{new} = I_r * K$$

$$I_r = \begin{cases} rate & rate \in [0.6, 1.5] \\ 1 & otherwise \end{cases} \quad (15)$$

4.2 BtPDR 的融合规则在智能手机跟踪过程中, 在 x_e 推导出 PDR 的定位结果和 x_e (BRPV 的定位结果) 之后, 下一个关键问题是如何融合 x_e 和 x_e 。如前所述, x_{ei} 很容易受到 RSS 波动的影响, x_e 处的票数反映了实际位置在某个区域内的概率; 票数越大, 真实位置在该地区的概率就越大。因此, 为了减少 RSS 波动的影响, BtPDR 丢弃了票数小于或等于阈值 β 的 x_e 。只有当票数 V_{of} BRPV 大于 β 时, x_{eis} 才被认为是可靠的。当 PDR 和 BRPV 的结果需要融合时, 融合基于:

$$x_{e_{融合}} = \frac{\frac{V_{PDR}}{V_{RSS} + V_{PDR}} \cdot x_{e_{PDR}} + \frac{V_{RSS}}{V_{RSS} + V_{PDR}} \cdot x_{e_{RSS}}}{\frac{V_{PDR}}{V_{RSS} + V_{PDR}} + \frac{V_{RSS}}{V_{RSS} + V_{PDR}}} \quad \text{如果 } V_{of} \geq \beta \cdot V_{PDR} \quad (13)$$

否则

其中 x_e 是最终的融合定位结果, V_{and} V_{are} 分别是 x_e 和 x_e 的概率投票的票数。从 (13) 中可以看出, x_e 的加权值是 x_e 和 x_e 的投票数。加权融合方法背后的思想是通过限制位置的校正范围来避免出现较大的误差, 使融合结果更加可靠。而且, BRPV 定位的初始位置只影响开始时的精度。BtPDR 对 PDR 和 BRPV 的定位结果进行多次融合后, 定位精度会收敛, 并且与初始位置无关。

4.3 BtPDR 的参数更新 BtPDR 融合了 PDR 和 BRPV 的定位结果后, 需要自适应更新 PDR 的步长。本研究假设 (x, y) 为第 k 次融合结果的坐标, (x_k, y_k) 为当前 PDR 的坐标。然后可以计算步长校正率:

$$\text{率} = \frac{\sqrt{(x_{k-1} - x_k)^2 + (y_{k-1} - y_k)^2}}{\sqrt{(x_{k-1} - x_{PDR})^2 + (y_{k-1} - y_{PDR})^2}} \quad (14)$$

当速率大于 0.6 且小于 1.5 时, 为有效修正。接下来, BtPDR 更新 (3) 中的 K , 如下所示:

$$I_r = \begin{cases} K_{新增功能} I_r * K & \text{率} \in [0.6, 1.5] \\ 1 & \text{否则} \end{cases} \quad (15)$$

5. Experimental results

5.1 The relationship between RSS and distance

This experiment was used to derive the relationship between RSS and distance in an actual scene. The experimental environment was an indoor rectangular corridor with a few obstacles.

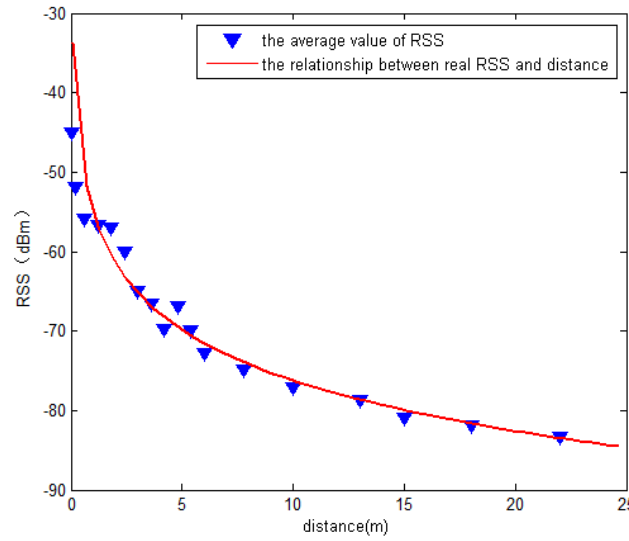


Fig. 9. The relationship between real RSS and distance

Bluetooth RSS data was collected by the smartphone at different distance from the Bluetooth beacon. At each location, 100 RSS acquisitions were carried out. Then, a Gaussian filter [29] was utilized to reduce the influence of small probability and big disturbance events on the overall measurement. The relationship between real RSS and distance by experiment is as shown in Fig. 9. The blue triangle in Fig. 9 is the average value of RSS after using a Gaussian filter and the red curve is the curve of (16).

Letting $d_0 = 1$, through the experiment the following values were determined: $P_{r,dB}(d_0) = -55\text{dBm}$, $n_p = 2.12$. The relationship between RSS and distance is then:

$$RSS = -55 - 10 \times 2.12 \times \log(d) + X \quad (16)$$

In the experiment, it was found that X obeys a Gaussian distribution of $N(0, \sigma^2)$ and σ varies with the received position of the smartphone. In the following experiment, for simplicity, $\sigma = 7$ was set and (16) was used to model all cases. It must be noted that 1) if only one measurement at one distance is made, the relationship between RSS and distance will not be as ideal as Fig. 9 and the error of the model in (16) will become bigger and 2) in the whole geographical space, the real relationship between RSS and distance does not comply with (16). Fortunately, experiment results show that the proposed BRPV and BtPDR can tolerate

5. 实验结果

5.1 RSS 与距离的关系本实验用于推导实际场景中 RSS 与距离的关系。实验环境为室内矩形走廊，有少量障碍物。

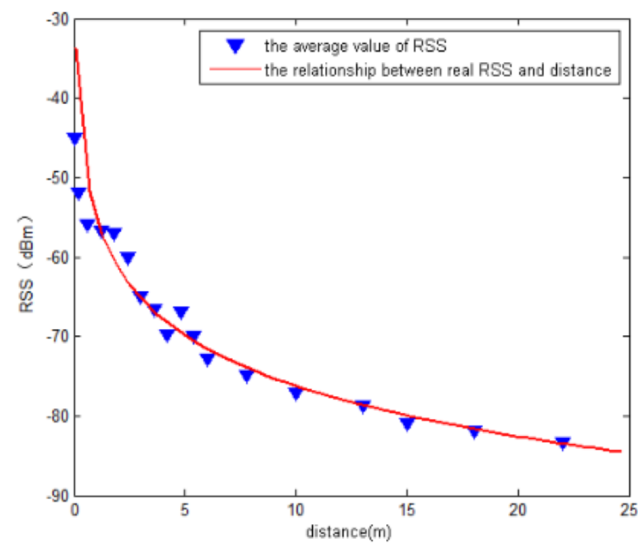


图 9.真实 RSS 与距离的关系

智能手机在与蓝牙信标不同距离处收集蓝牙 RSS 数据。在每个位置，进行了 100 次 RSS 采集。然后，利用高斯滤波器[29]来减少小概率和大干扰事件对整体测量的影响。实验中真实 RSS 与距离的关系如图 9 所示。图 9 中的蓝色三角形是使用高斯滤波器后 RSS 的平均值，红色曲线是 (16) 的曲线。

设 $d=1$ ，通过实验确定了以下值： $P(d=-55\text{dBm}, n=2.12)$ 。RSS 与距离之间的关系为：

$$\text{RSS} \approx -55 - 10 \times 12 \log(d + X) \tag{16}$$

在实验中发现， X 服从 $N(0, \sigma)$ 的高斯分布，并且 σ 随智能手机的接收位置而变化。在下面的实验中，为简单起见，设置了 $\sigma=7$ ，并使用 (16) 对所有情况进行建模。必须注意的是，1) 如果在一个距离上只进行一次测量，RSS 与距离之间的关系将不会像图 9 那样理想，并且 (16) 中模型的误差会变大，2) 在整个地理空间中，RSS 与距离之间的真实关系不符合 (16)。幸运的是，实验结果表明，所提出的 BRPV 和 BtPDR 可以容忍

the suggested model errors, yielding satisfactory localization results.

5.2 The performance analysis of BRPV

To prove the localization accuracy of BRPV, simulations were carried out using MATLAB software. The simulation parameters are as follows: the number of total nodes is 104, including 4 Bluetooth beacons and 100 unknown nodes; the positions of the beacons are fixed and their coordinates are (6, 6), (0, 12), (6, 18), and (0, 24); the unknown nodes are uniformly distributed, in a fixed position, in a $6\text{m} \times 25\text{m}$ rectangular region and each unknown node acquires 10 RSS for each beacon node. The experimental results are the averaged results of 1000 experiments.

5.2.1 The area division of BRPV

This section analyzes the impact of the number of areas of BRPV. The plane is divided into three, five, seven, and nine areas, and the number of votes is assigned according to the probability that the smartphone falls into these areas. Simulation results as shown in [Table 1](#), where BRPV uses the full search method in section 3.3. From [Table 1](#), it can be determined that having three or nine areas is inappropriate. Compared to five areas, seven areas can improve localization accuracy by 1.76% at the cost of increasing CPU time by 26.4%. Therefore, considering the localization accuracy and complexity, BRPV employed five areas in the following experiments.

Table 1. The impact of area division on algorithm performance

The number of areas	CPU time per node (ms)	Absolute localization error (m)
3	50.1	6.732
5	70.3	1.534
7	88.9	1.507
9	105.6	1.528

5.2.2 Compared BRPV to other algorithms

In this section, BRPV is compared with the traditional least squares based multilateral positioning (LSMP) and fingerprint positioning (FP) methods. To do so, a fingerprint reference point was deployed every two meters, creating 52 reference points. For each reference point, 50 fingerprint acquisitions were carried out and the averaged fingerprint was used as the characteristic fingerprint of the reference point. The comparison results under different noise standard deviations are shown in [Fig. 10](#).

建议的模型误差, 产生了令人满意的定位结果。

5.2 BRPV 的性能分析为了证明 BRPV 的定位精度, 使用 MATLAB 软件进行了仿真。仿真参数如下: 总节点数为 104 个, 其中蓝牙信标 4 个, 未知节点 100 个;信标位置固定, 坐标为 (6, 6)、(0, 12)、(6, 18)、(0, 24) ;未知节点在固定位置均匀分布在 6 m×25 m 的矩形区域, 每个未知节点为每个信标节点获取 10 个 RSS。实验结果为 1000 次实验的平均结果。

5.2.1 BRPV 的区域划分本节分析了 BRPV 区域数量的影响。将平面分为三个、五个、七个和九个区域, 根据智能手机落入这些区域的概率分配票数。仿真结果如表 1 所示, 其中 BRPV 使用了第 3.3 节中的全搜索方法。从表 1 可以确定, 有三个或九个区域是不合适的。与五个区域相比, 七个区域可以提高 1.76%的定位精度, 代价是 CPU 时间增加了 26.4%。因此, 考虑到定位精度和复杂性, BRPV 在以下实验中采用了五个区域。

表 1.面积划分对算法性能的影响		
区域数	每个节点的 CPU 时间 (ms)	绝对定位误差 (m)
3	50.1	6.732
5	70.3	1.534
7	88.9	1.507
9	105.6	1.528

5.2.2 BRPV 与其他算法的比较在本节中, 将 BRPV 与传统的基于最小二乘法的多边定位 (LSMP) 和指纹定位 (FP) 方法进行了比较。为此, 每两米部署一个指纹参考点, 创建 52 个参考点。对于每个参考点, 进行 50 次指纹采集, 并将平均指纹用作参考点的特征指纹。不同噪声标准差下的比较结果如图 10 所示。

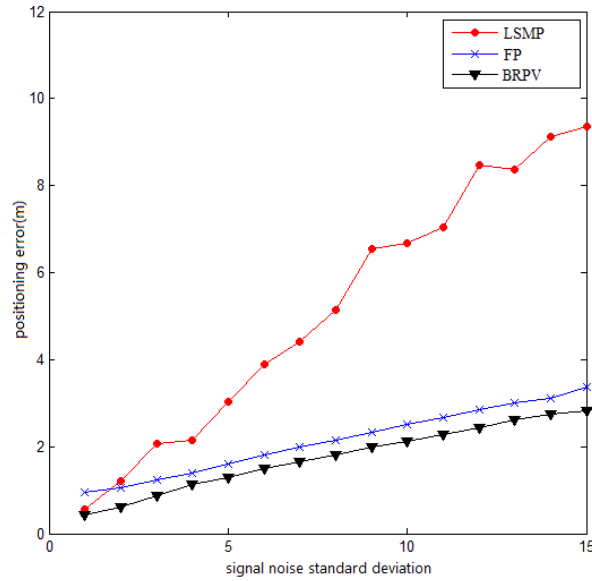


Fig. 10. Comparison of BRPV to LSMP and FP

From **Fig. 10** it can be determined that 1) BRPV and FP have similar tolerance to noise, which is superior to the LSMP and 2) the localization accuracy of BRPV is slightly better than that of the FP and much better than that of the LSMP.

5.3 The performance analysis of the BtPDR

In this section, the theoretical simulation and test experiments of BtPDR in the real scene are described. The comparison algorithms include the basic PDR in [14], the proposed BRPV, the LSMP in [6], and the Kalman filter-based fingerprint positioning (KFFP) algorithm in [12].

5.3.1 Theoretical simulation

The theoretical simulation parameters in this paper are as follows: 1) four Bluetooth beacons in a localization scene were deployed and their coordinates were (6, 6), (0, 12), (6, 18) and (0, 24); 2) the actual walking route was a $4.8\text{m} \times 22.2\text{m}$ rectangle and 3) considering that the number of received RSS samples in each real step was very low, in the simulation, each step only received one RSS sample. Moreover, the walking model of PDR is:

$$\begin{aligned} L &= L_0 + A + X_{\sigma_1} \\ \theta &= \theta_0 + B + X_{\sigma_2} \end{aligned} \quad (17)$$

Where L_0 and θ_0 is the real length and heading of each step and A and B are constants used to simulate the fixed error caused by sensor accuracy limits, which were set to $A = 0.1\text{m}$, $B = 3^\circ$ in simulation. X_{σ_1} and X_{σ_2} are zero mean Gaussian noises; σ_1 is the standard deviation of the step length, which was set to $\sigma_1 = 0.1$ and σ_2 is the standard deviation of heading, which was

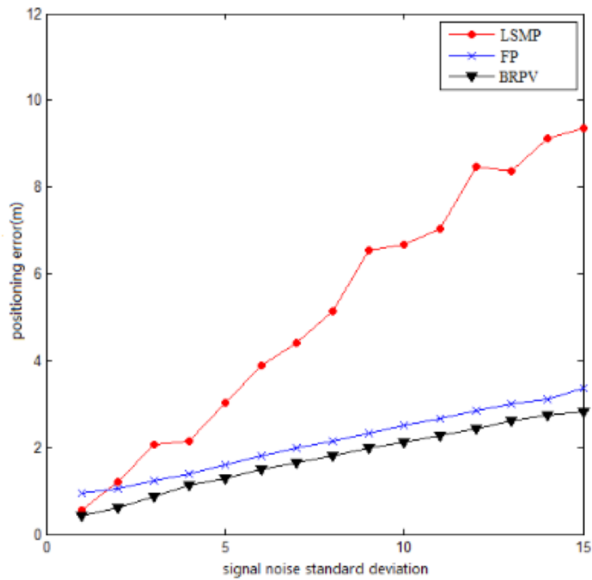


图 10.BRPV 与 LSMP 和 FP 的比较

从图 10 可以确定, 1) BRPV 和 FP 对噪声的容忍度相似, 优于 LSMP, 2) BRPV 的定位精度略优于 FP, 远优于 LSMP。

5.3 BtPDR 的性能分析本节介绍了 BtPDR 在真实场景中的理论仿真和测试实验。比较算法包括 [14] 中的基本 PDR、[6] 中提出的 BRPV、LSMP 和 [12] 中的基于卡尔曼滤波的指纹定位 (KFFP) 算法。

5.3.1 理论仿真本文的理论仿真参数如下: 1) 在定位场景中部署了 4 个蓝牙信标, 其坐标为 (6,6), (0,12), (6,18) 和 (0,24); 2) 实际行走路线为 4.8m×22.2m 矩形, 3) 考虑到每个实际步骤中接收到的 RSS 样本数量非常少, 在仿真中, 每个步骤只接收到一个 RSS 样本。此外, PDR 的行走模型为:

$$L = L_0 + A X_{s1}$$
$$\theta = \theta_0 + B + X_{s2}$$

(17)

式中, L 和 θ 是每步的实际长度和航向, A 和 B 是用于模拟传感器精度极限引起的固定误差的常数, 在仿真中设置为 $A = 0.1\text{m}$, $B = 3^\circ$ 。 X_1 和 X_2 为零平均高斯噪声; σ_1 是步长的标准差, 设置为 $\sigma_1 = 0.1$, σ_2 是航向的标准差, 即

set to $\sigma^2 = 3.87$.

Since the simulation scene only had four Bluetooth beacons, according to Fig. 4 we can see that the range of votes in each region should be [4, 24]. Therefore, to ensure that the fusion results of BRPV and PDR are reliable, $\beta = 18$ was set in the simulation. The fusion condition is that the number of votes of BRPV is larger than 18; otherwise, the localization results of BRPV are discarded and the localization results of PDR used as the final localization results.

Under the conditions of the theoretical experiment, the comparison results of different localization algorithms with different RSS noise errors are shown in Table 2 and Table 3.

Table 2. Comparison of algorithm performance ($\sigma^2 = 20$)

Algorithm	positioning error/m	90% positioning error/m	accuracy within 1m /%	accuracy within 2m /%	accuracy within 3m /%	accuracy within 5m /%
PDR	3.17	4.79	21	38	56	94
BRPV	2.06	4.97	26	68	79	90
LSMP	4.96	9.76	16	32	40	61
KFFP	1.87	3.78	34	62	73	95
BtPDR	1.13	2.62	50	85	97	100

Table 3. Comparison of algorithm performance ($\sigma^2 = 50$)

Algorithm	positioning error/m	90% positioning error/m	accuracy within 1m /%	accuracy within 2m /%	accuracy within 3m /%	accuracy within 5m /%
PDR	3.28	5.29	18	33	51	84
BRPV	3.21	6.81	16	36	58	76
LSMP	8.56	15.65	8	16	26	54
KFFP	2.55	5.65	22	40	61	84
BtPDR	1.66	2.72	35	66	97	100

Table 2 and Table 3 show that BtPDR's positioning accuracy is the best because it has the smallest positioning error in these algorithms. Compared with PDR, BRPV, LSMP and KFFP, BtPDR has improved positioning accuracy by 64.3%, 45.1%, 77.2% and 39.6%, respectively, in the case of a noise variance of 20, and has improved positioning accuracy by 49.4%, 48.3%, 80.6% and 34.9%, respectively, in the case of a noise variance of 50. With the increased noise variance of Bluetooth RSS, 1) the positioning accuracy of BRPV, LSMP and KFFP clearly became worse because they are related to Bluetooth RSS; 2) the positioning accuracy of PDR became small because PDR is unrelated to Bluetooth RSS and 3) the positioning error of BtPDR became large due to the increase of the positioning error of the BRPV. Note that the reason for the positioning error of BRPV and LSMP in Table 2 and Table 3 being significantly larger than the results in Table 1 is that each set step only

设置为 $\sigma^2 = 3.87$ 。

由于仿真场景只有四个蓝牙信标, 根据图 4 我们可以看到每个区域的投票范围应该为[4,24]。因此, 为了保证 BRPV 和 PDR 的融合结果可靠, 在仿真中设置了 $\beta=18$ 。融合条件是 BRPV 的投票数大于 18;否则, 将丢弃 BRPV 的定位结果, 并将 PDR 的定位结果作为最终的定位结果。

在理论实验条件下, 不同 RSS 噪声误差的不同定位算法的对比结果如表 2 和表 3 所示。

表 2.算法性能比较 ($\sigma^2 = 20$)						
算法	定位 误差/月	90% 定位 误差/月	准确性 1m 以内 /%	准确性 2 米以内 /%	准确性 3m 以内 /%	准确性 5 米以内 /%
人民共和国	1.17	4.79	21	38	56	94
BRPV 的	2.06	4.97	26	68	79	90
LSMP 的	4.96	9.76	16	32	40	61
KFFP	1.87	3.78	34	62	73	95
BtPDR	1.13	2.62	50	85	97	100

表 3.算法性能比较 ($\sigma^2 = 50$)						
算法	定位 误差/月	90% 定位 误差/月	准确性 1m 以内 /%	准确性 2 米以内 /%	准确性 3m 以内 /%	准确性 5 米以内 /%
人民共和国	1.28	5.29	18	33	51	84
BRPV 的	3.21	6.81	16	36	58	76
LSMP 的	8.56	15.65	8	16	26	54
KFFP	2.55	5.65	22	40	61	84
BtPDR	1.66	2.72	35	66	97	100

表 2 和表 3 表明, BtPDR 的定位精度是最好的, 因为它在这些算法中的定位误差最小。与 PDR、BRPV、LSMP 和 KFFP 相比, 在噪声方差为 20 的情况下, BtPDR 的定位精度分别提高了 64.3%、45.1%、77.2%和 39.6%, 在噪声方差为 50 的情况下, 定位精度分别提高了 49.4%、48.3%、80.6%和 34.9%。随着蓝牙 RSS 噪声方差的增大, 1) BRPV、LSMP 和 KFFP 的定位精度明显变差, 因为它们与蓝牙 RSS 相关;2) PDR 的定位精度变小, 因为 PDR 与蓝牙 RSS 无关, 3) BtPDR 的定位误差变大, 因为 BPRV 的定位误差增加。请注意, 表 2 和表 3 中 BRPV 和 LSMP 的定位误差明显大于表 1 中的结果的原因是每个设置步骤仅

received one RSS sample in the simulation, causing the RSS to have stronger fluctuation, thus increasing the positioning error. By contrast, each step in [Table 1](#) received ten RSS samples and an average of RSS samples could be calculated to resist the RSS fluctuation.

5.3.2 Actual location experiments

Experiments were conducted in three actual scenarios including a separate indoor corridor, a separate office and a indoor corridor + office. A separate indoor corridor was used to test the performance of the localization algorithm in an ideal environment with few obstacles. A separate office was used to test the performance of the localization algorithm in a complex environment with lots of obstacles. A corridor + office environment was used to test the performance of the localization algorithm in a general environment.

The experimental field was a $42.2\text{m} \times 21\text{m}$ indoor environment as shown in [Fig. 11](#). There were 12 Bluetooth beacons in the experiment field. The red stars in [Fig. 11](#) represent the location of the Bluetooth beacons. The BRPV only used the 4 largest Bluetooth beacon RSS to locate the smartphone, hence, the range of the number of votes for each voting area in BRPV was $[4, 24]$. To enhance the reliability of BtPDR, $\beta = 20$ was set in the actual scene. The Bluetooth beacons used a CC2541 Bluetooth 4.0 module, produced by Texas Instruments, with a radio transmission power of 0dBm and a launch rate of five times per second. The smartphone was a MX4 MeiZu, which has a built-in accelerometer, direction sensor, Bluetooth 4.0 chip and Android 4.4 OS.

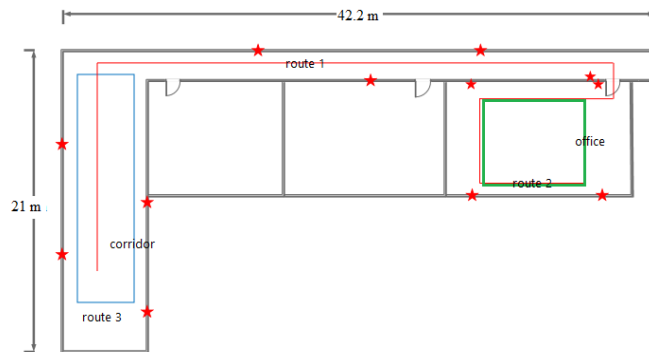


Fig. 11. Experimental field map

To verify the robustness of BtPDR, three experimenters were invited to carry out the experiment. Each experimenter carried out four experiments for each experiment environment. Therefore, 36 experiments in total were carried out for three experimental environments. Since the sex, height, frequency and handheld mobile phone habits of each experimenter was different, the results can be used to test the robustness of BtPDR for different experimenters.

请注意, 表 2 和表 3 中 BRPV 和 LSMP 的定位误差明显大于表 1 的结果的原因是, 每个设置步骤在仿真中只接收到一个 RSS 样本, 导致 RSS 具有更强的波动, 从而增加了定位误差。相比之下, 表 1 中的每个步骤接收 10 个 RSS 样本, 可以计算出 RSS 样本的平均值来抵抗 RSS 波动。

5.3.2 实际位置实验 在单独的室内走廊、单独的办公室和室内走廊+办公室的三个实际场景下进行了实验。使用单独的室内走廊来测试定位算法在障碍物较少的理想环境中的性能。使用单独的办公室来测试定位算法在障碍物较多的复杂环境中的性能。使用走廊+办公环境来测试定位算法在一般环境中的性能。

实验场为 $42.2\text{m} \times 21\text{m}$ 的室内环境, 如图 11 所示。实验场内有 12 个蓝牙信标。图 11 中的红星表示蓝牙信标的位置。BRPV 仅使用 4 个最大的蓝牙信标 RSS 来定位智能手机, 因此, BRPV 中每个投票区的票数范围为 $[4, 24]$ 。为了增强 BtPDR 的可靠性, 在实际场景中设置了 $\beta = 20$ 。蓝牙信标采用德州仪器 (TI) 生产的 CC2541 蓝牙 4.0 模块, 无线电发射功率为 0dBm, 发射速率为每秒 5 次。智能手机是 MX4 魅族, 内置加速度计、方向传感器、蓝牙 4.0 芯片和 Android 4.4 作系统。

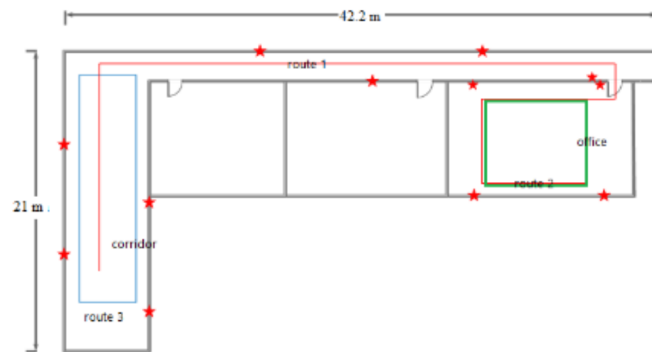


图 11. 实验场图

为了验证 BtPDR 的鲁棒性, 邀请了 3 名实验者进行实验。每个实验者针对每个实验环境进行了 4 次实验。因此, 在 3 个实验环境中总共进行了 36 次实验。由于每个实验者的性别、身高、频率和手持手机习惯不同, 因此该结果可用于测试 BtPDR 对不同实验者的鲁棒性。

a) indoor corridor + office

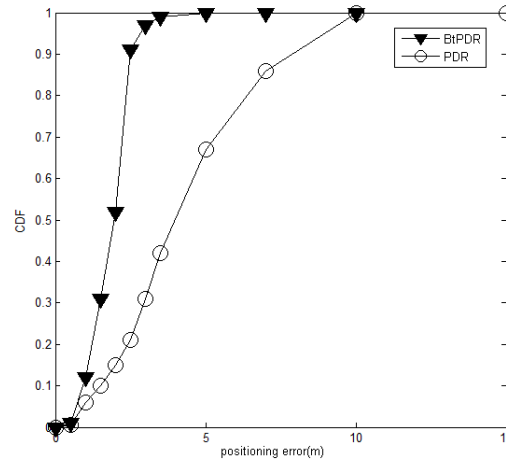
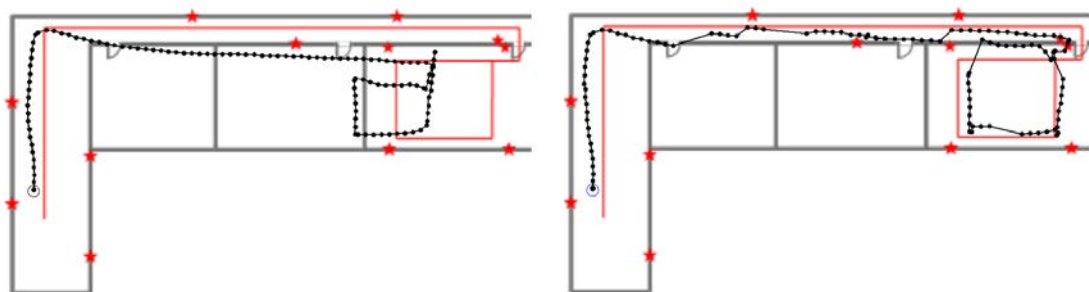


Fig. 12. The CDF of the positioning error

Experimental environment I was indoor corridor + office. The actual walking trajectory is represented by route 1 in **Fig. 11**. Route 1 had a distance of 82.7m and passed indoor corridors and offices. The average absolute positioning errors of PDR and BtPDR were 4.23m and 1.86m, respectively, and the cumulative distribution function (CDF) of the positioning error is shown in **Fig. 12**. Therefore, in the indoor corridor + office environment, BtPDR improved the positioning accuracy by up to 56%, compared to PDR.

Fig. 13 compares location route results. As shown in **Fig. 13(a)**, since the traditional PDR cannot accurately estimate or adaptively adjust the step size, the difference between positioning trajectory and actual trajectory is large. However, BtPDR can adaptively fuse the localization results of PDR and BRPV, adjusting the parameters of PDR when the localization results of BRPV are credible. Therefore, the positioning trajectory of BtPDR in **Fig. 13(b)** is clearly closer to the actual trajectory than that of PDR.



(a) The positioning trajectory of PDR

(b) The positioning trajectory of BtPDR

Fig. 13. Comparison of location route

a) 室内走廊+办公室

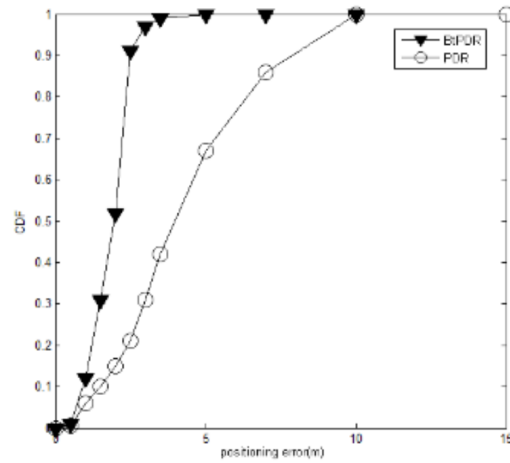
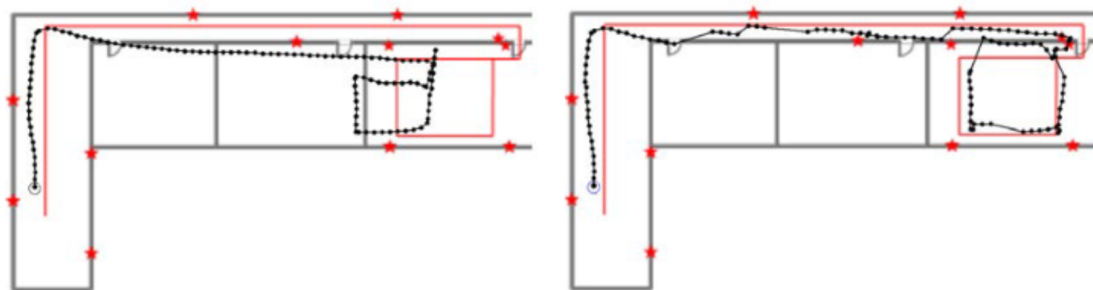


图 12.定位误差的 CDF

实验环境 I 为室内走廊+办公室。实际行走轨迹由图 11 中的路线 1 表示。路线 1 的距离为 82.7m, 经过室内走廊和办公室。PDR 和 BtPDR 的平均绝对定位误差分别为 4.23m 和 1.86m, 定位误差的累积分布函数 (CDF) 如图 12 所示。因此, 在室内走廊+办公环境中, BtPDR 比 PDR 提高了 56% 的定位精度。

图 13 对比了定位路径结果。如图 13 (a) 所示, 由于传统 PDR 无法准确估计或自适应调整步长, 因此定位轨迹与实际轨迹的差异较大。然而, BtPDR 可以自适应地融合 PDR 和 BRPV 的定位结果, 在 BRPV 的定位结果可信时调整 PDR 的参数。因此, 图 13 (b) 中 BtPDR 的定位轨迹明显比 PDR 更接近实际轨迹。



(a) 人民民主共和国的定位轨迹

(b) BtPDR 的定位轨迹

图 13.位置路线对比

Fig. 14 compares cumulative error results. It can be seen that the cumulative error of PDR gradually increases with an increased number of steps and the error cannot be self-corrected. But BtPDR can correct the length of a forward step and the cumulative error will not increase with the increase of the number of steps.

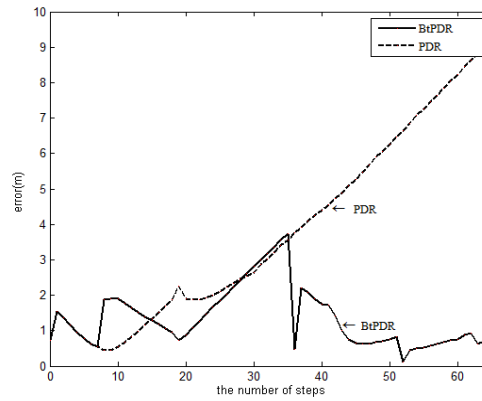


Fig. 14. Cumulative error distribution

b) Separate office environment

Experimental environment II consisted of only an office environment. The walking route is represented by route 2 in **Fig. 11**. Each experimenter walked along the green path two times for a total of 53.6m. The average absolute positioning errors of PDR and BtPDR were 2.59m and 1.94m, respectively, and the positioning error CDF is shown in **Fig. 15**. Location route comparison results are shown in **Fig. 16**. It can be seen that, in the office environment, BtPDR improved the positioning accuracy by up to 25.1% compared to PDR. The positioning error of PDR in experiment environment II is clearly smaller than that of PDR in experiment environment I. The reasons for this are: 1) the walking route of experiment II was shorter than that of experiment I and PDR error increases as the walking path increases and 2) the walking route of experiment II was a complete circular route and the positioning error of PDR was offset due to the circular shape of route.

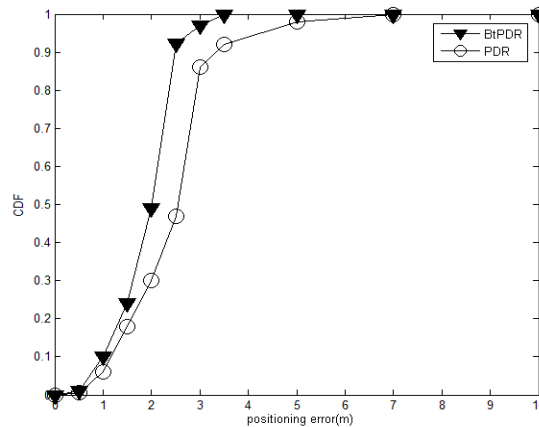


Fig. 15. The cumulative distribution function of the positioning error

图 14 对比了累积误差结果。可以看出, PDR 的累积误差随着步数的增加而逐渐增大, 误差无法自我纠正。但 BtPDR 可以纠正一个前进步的长度, 累积误差不会随着步数的增加而增加。

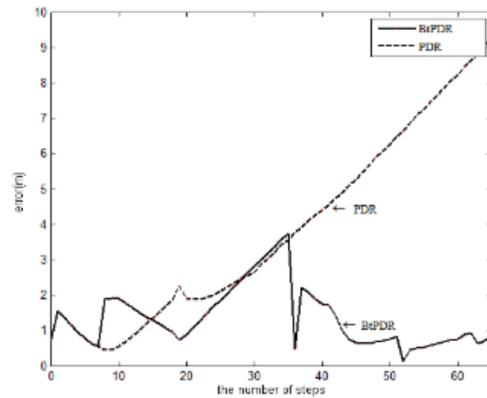


图 14.累积误差分布

b) 独立的办公环境实验环境 II 仅由一个办公环境组成。步行路线由图 11 中的路线 2 表示。每个实验者沿着绿色路径行走两次, 总共 53.6m。PDR 和 BtPDR 的平均绝对定位误差分别为 2.59m 和 1.94m, 定位误差 CDF 如图 15 所示。位置路线比较结果如图 16 所示。可以看出, 在办公环境中, BtPDR 比 PDR 提高了 25.1%的定位精度。实验环境 II 中 PDR 的定位误差明显小于实验环境 I 中的 PDR。其原因在于: 1) 实验 II 的步行路线比实验 I 的步行路线短, PDR 误差随着步行路径的增加而增大, 2) 实验 II 的步行路线是一条完整的环形路线, PDR 的定位误差因路线的圆形形状而被抵消。

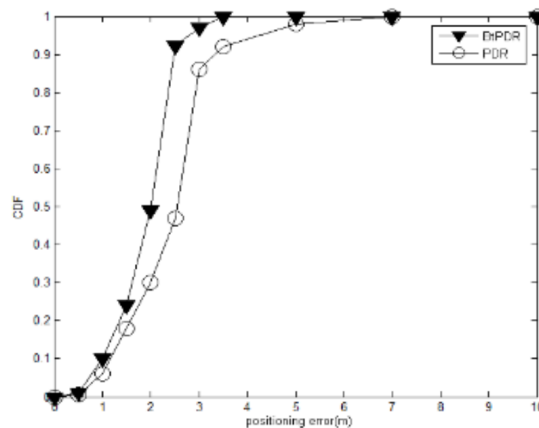


图 15.定位误差的累积分布函数

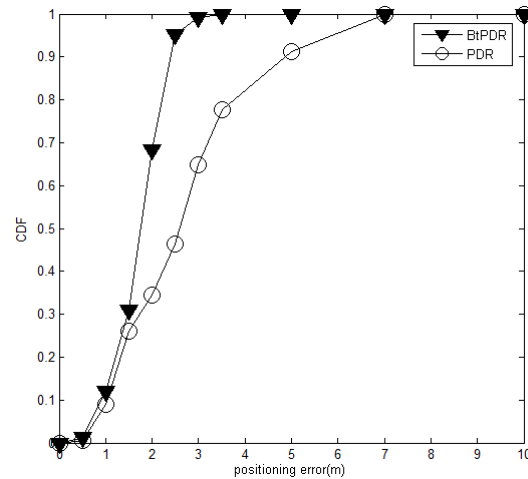


(a) positioning trajectory of PDR

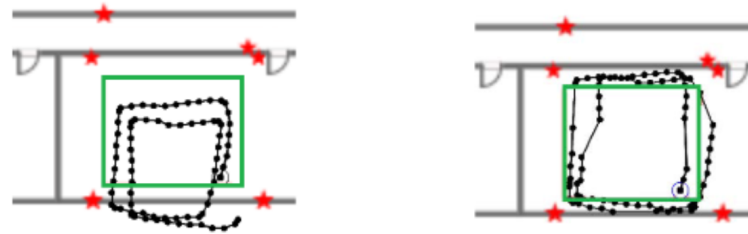
(b) positioning trajectory of BtPDR

Fig. 16. Comparison of location routes

c) Separate corridor environment

**Fig. 17.** The CDF of the positioning error in the corridor environment

Experimental environment III consisted of only a corridor environment. The walking route is represented by route 3 in **Fig. 11**. Each experimenter walked along the blue path two times for a total of 69.6m. The average absolute positioning errors of PDR and BtPDR were 2.97m and 1.58m, respectively, and the positioning error CDF is shown in **Fig. 17**. Location route comparison results are shown in **Fig. 18**. It can be seen that, in the corridor environment, BtPDR improved the positioning accuracy by up to 46.8% compared to PDR. In addition, because the corridor environment is an ideal indoor environment with few obstacles, BRPV had more accurate localization results causing localization performance improvement of BtPDR. Therefore, BtPDR in the corridor environment has the best positioning accuracy.

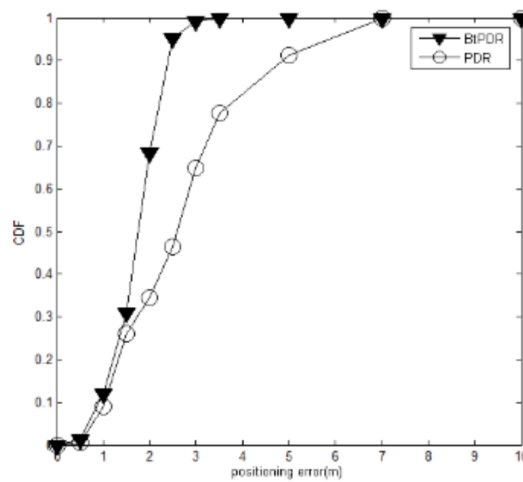


(a) PDR 的定位轨迹

(b) BtPDR 的定位轨迹

图 16.位置路线对比

c) 独立楼道环境图 17.楼道环境中定位误差的 CDF



实验环境 III 仅由走廊环境组成。步行路线由图 11 中的路线 3 表示。每个实验者沿着蓝色路径行走两次, 总共 69.6m。PDR 和 BtPDR 的平均绝对定位误差分别为 2.97m 和 1.58m, 定位误差 CDF 如图 17 所示。位置路径对比结果如图 18 所示。可以看出, 在廊道环境中, BtPDR 比 PDR 提高了 46.8% 的定位精度。此外, 由于廊道环境是障碍物较少的理想室内环境, BRPV 具有更准确的定位结果, 从而提高了 BtPDR 的定位性能。因此, 廊道环境中的 BtPDR 具有最好的定位精度。

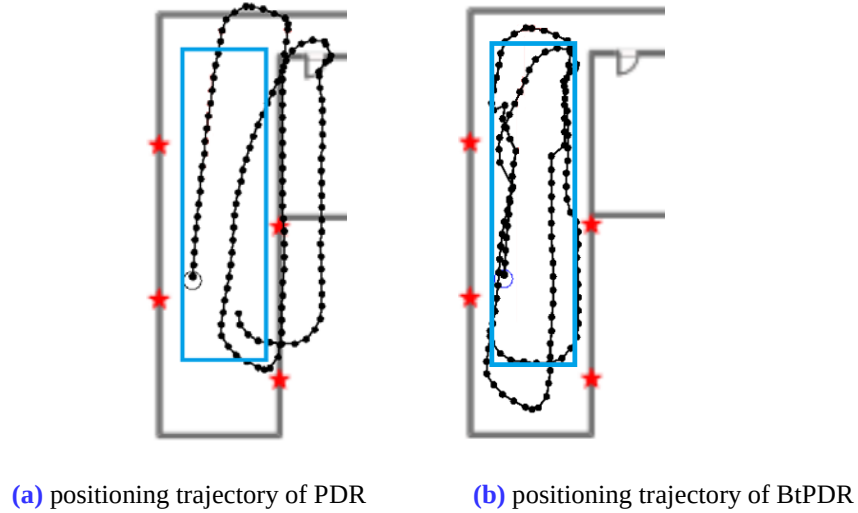


Fig. 18. Comparison of location routes in the corridor environment

5.3.3 The Latency of BtPDR

Latency is another key indicator of the positioning system in practice. An average person walks about 95~125 steps per minute, meaning their speed is about 480~630ms per step. Through the experiment, we found that BtPDR requires about 30ms of processing time for each step. Therefore, this study argues that, in practical application, the latency of BtPDR meets real-time positioning.

5.4 Compared with other indoor fusion localization algorithms

Because Bluetooth and PDR-based fusion localization methods are seldom seen in the literature and most fusion localization methods are based on WLAN and PDR, the indoor localization algorithms in [16-18] were chosen for comparison. These algorithms were chosen because their experimental environments are like this study's experimental environment and they are relatively new fusion algorithms. The work in [16] and [17] are fusion localization algorithms based on PDR and WLAN fingerprints. The experiment settings in [16] included a HTC Hero with Android 1.5 OS mobile terminal; 24 WLAN beacons in the positioning region; a fingerprint reference point deployed every three meters creating 183 reference points; and for each reference point, 100 fingerprint acquisitions were carried out. In [16], the average positioning errors of three test paths were 2.76m, 2.57m and 2.81m, which is worse than the results of BtPDR. The experiment settings in [17] included a Samsung Galaxy S2 with Android 2.2 OS mobile terminal; 10 WLAN beacons in the positioning region; a fingerprint reference point deployed every two meters creating 72 reference points; and for each reference point, 500 fingerprint acquisitions were carried out.

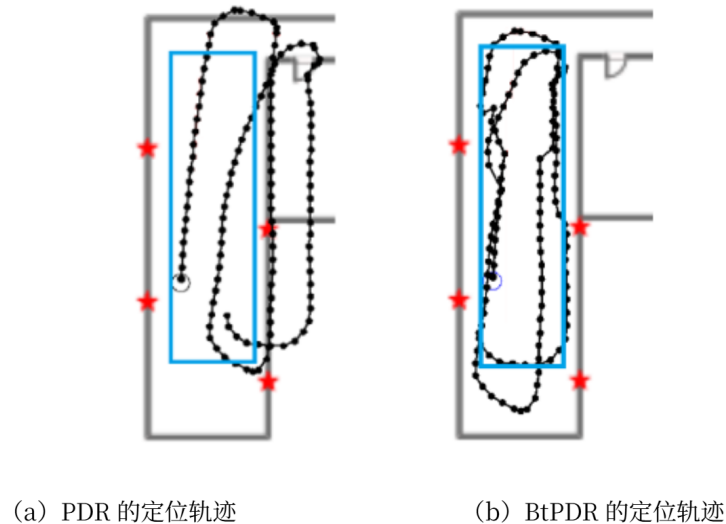


图 18. 廊道环境中的定位路线对比

5.3.3 BtPDR 的延迟延迟是定位系统在实践中另一个关键指标。一个普通人每分钟步行约 95~125 步, 这意味着他们的速度约为每步 480~630ms。通过实验, 我们发现 BtPDR 每步需要大约 30ms 的处理时间。因此, 本研究认为, 在实际应用中, BtPDR 的延迟满足了实时定位。

5.4 与其他室内融合定位算法的比较由于基于蓝牙和 PDR 的融合定位方法在文献中很少见, 而且大多数融合定位方法都是基于 WLAN 和 PDR 的, 因此选择了[16-18]中的室内定位算法进行比较。选择这些算法是因为它们的实验环境类似于本研究的实验环境, 并且是相对较新的融合算法。[16]和[17]中的工作是基于 PDR 和 WLAN 指纹的融合定位算法。[16]中的实验设置包括一个带有 Android 1.5 OS 移动终端的 HTC Hero; 定位区域有 24 个 WLAN 信标; 每三米部署一个指纹参考点, 创建 183 个参考点; 对于每个参考点, 进行了 100 次指纹采集。在[16]中, 3 条测试路径的平均定位误差分别为 2.76m、2.57m 和 2.81m, 比 BtPDR 的结果差。[17]中的实验设置包括一台搭载 Android 2.2 OS 移动终端的三星 Galaxy S2; 定位区域有 10 个 WLAN 信标; 每两米部署一个指纹参考点, 创建 72 个参考点; 对于每个参考点, 进行了 500 次指纹采集。

In [17], the experimental results showed that 90% of the positioning error was less than 4m and the average positioning error was 1.9m. The average positioning error in [17] is like that of BtPDR, but a 90% node localization error is less than the results of BtPDR. The work in [18] used PDR and map information to realize fusion localization. The experiment settings in [18] included a sensor worn on the waist and map information needed to detect the correctness of trajectory. The average positioning error of the two experiments was 1.4m and 1.7m, respectively. The reasons that the experimental results in [18] are superior to BtPDR are: compared to a sensor held in the hand, the sensor worn on the waist can reduce the volatility of the collected data and the experiment used extra map information and a complex particle filter to refine PDR results.

In summary, compared to previous literature, the proposed BtPDR not only provides accurate indoor localization but also eliminates the work for acquisition of offline fingerprint or map information. Moreover, to validate the robustness of BtPDR, three testers were chosen to carry out the experiment in three different experimental scenarios. The work in [16-18] did not mention conducting a robustness experiment and the indoor environment was an ideal corridor environment. Therefore, this study suggests that the robustness of BtPDR is superior to that of the algorithms in [16-18].

6. Conclusion

This study presented an easy to implement approach, BtPDR, for indoor localization and trajectory tracking that combines Bluetooth RSS with PDR. First, the initial position of a smartphone was localized with BRPV. Second, while the smartphone was moving, it was tracked with PDR and BRPV at the same time. Last, localization results of PDR and BRPV were fused and parameters of PDR were adaptively modified. Experiment results show that the positioning error of BtPDR is less than 2m, which is significantly better than that of PDR and BRPV. Moreover, experiment results also show that, compared to the WLAN fingerprint and PDR-based fusion indoor localization approaches, BtPDR has better positioning accuracy and does not need the offline workload of a fingerprint algorithm. Therefore, the proposed BtPDR is accurate, stable and practical. This study is not without limitations. Due to the limitation of experimental conditions, localization experiments were not carried out in a wide range of indoor areas; the RSS error due to the shadow effect of body occlusion was not considered and considering the recent data precision enhancement techniques of IMU sensors to further improve the localization accuracy of PDR would be beneficial. These shortcomings will be addressed in future work.

在[17]中, 实验结果表明, 90%的定位误差小于 4m, 平均定位误差为 1.9m。[17]中的平均定位误差与 BtPDR 相似, 但 90%的节点定位误差小于 BtPDR 的结果。[18]中的工作使用 PDR 和地图信息实现融合定位。[18]中的实验设置包括佩戴在腰部的传感器和检测轨迹正确性所需的地图信息。两个实验的平均定位误差分别为 1.4 m 和 1.7 m。[18]中的实验结果优于 BtPDR 的原因是: 与手中的传感器相比, 佩戴在腰部的传感器可以降低采集数据的波动性, 并且实验使用了额外的地图信息和复杂的粒子滤波器来细化 PDR 结果。

综上所述, 与以往的文献相比, 所提出的 BtPDR 不仅提供了准确的室内定位, 而且省去了获取离线指纹或地图信息的工作。此外, 为了验证 BtPDR 的鲁棒性, 选择了三个测试人员在三种不同的实验场景下进行实验。[16-18]中的工作没有提到进行鲁棒性实验, 室内环境是理想的走廊环境。因此, 本研究表明 BtPDR 的鲁棒性优于[16-18]中的算法。

六、结论

本研究提出了一种易于实现的室内定位和轨迹跟踪方法 BtPDR, 该方法将蓝牙 RSS 与 PDR 相结合。首先, 使用 BRPV 定位智能手机的初始位置。其次, 在智能手机移动的同时, 使用 PDR 和 BRPV 进行跟踪。最后, 融合 PDR 和 BRPV 的定位结果, 并自适应修改 PDR 的参数。实验结果表明, BtPDR 的定位误差小于 2m, 明显优于 PDR 和 BRPV。此外, 实验结果还表明, 与 WLAN 指纹和基于 PDR 的融合室内定位方法相比, BtPDR 具有更好的定位精度, 并且不需要指纹算法的离线工作负载。因此, 所提出的 BtPDR 具有准确、稳定和实用的特点。这项研究并非没有局限性。由于实验条件的限制, 没有在较广泛的室内区域进行定位实验;没有考虑身体遮挡阴影效应导致的 RSS 误差, 考虑 IMU 传感器的最新数据精度增强技术, 进一步提高 PDR 的定位精度是有益的。这些缺点将在未来的工作中得到解决。

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